INSTRUCTOR SOLUTION

Habib University

Introduction to Probability and Statistics – EE 354/CE 361/MATH 310 – Spring 2026

Instructor – Dr. Aamir Hasan - Section L4

Time = 20 Minutes **Quiz No 09** [20 points]

Provide all your answers on the answer sheet. Present your final answers and reasoning clearly.

Most of the total score for the questions will be awarded on your reasoning.

CLO No 04 - Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables. (Cog - 3)

Question 1 [14 points] [Conditional Probability Density Function, 2-dimensional]

Random Variables X and **Y** are jointly continuous and has the following probability density function $f_{X,Y}(x,y)$:

$$f_{X,Y}(x,y) = \begin{cases} c(x+y), & 0<x<1,\ 0<y<1 \\ 0, & else \end{cases}$$

Note – provide necessary justifications and applicable regions where probability density functions are applicable.

Part a [02 points] Determine the value of the constant c.

Part b [04 points] Find the probability density function, $f_Y(y)$.

Part c [10 points] Find the conditional probability density function, $f_{X|Y}(x|y)$ and plot $f_{X|Y}\left(x \,\middle|\, y=\frac{1}{2}\right)$.

Part d [04 points] Find the Probability, $P\left(X \le \frac{1}{2} \middle| Y = \frac{1}{2}\right)$.

(a) $\int_x \int_y f_{X,Y}(x,y)\,dy\,dx = 1 \Rightarrow \int_x \left[c\left(xy + \frac{y^2}{2}\right)\right]_0^1 dx = 1$ — 1

$\Rightarrow \int_x c\left(x + \frac{1}{2}\right) dx = c\left[\frac{x^2}{2} + \frac{1}{2}x\right]_0^1 = 1$

$= c \cdot 1 = 1$

$\Rightarrow c = 1$ — 1

$$f_{X,Y} = \begin{cases} x+y & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0 & else \end{cases}$$

(b) $f_Y(y) = \int_0^1 f_{X,Y}(x,y)\,dx$ — 1

$= \frac{x^2}{2} + yx \Big|_0^1$ — 1

$f_Y(y) = y + \frac{1}{2} \quad 0 \le y \le 1$ — 2

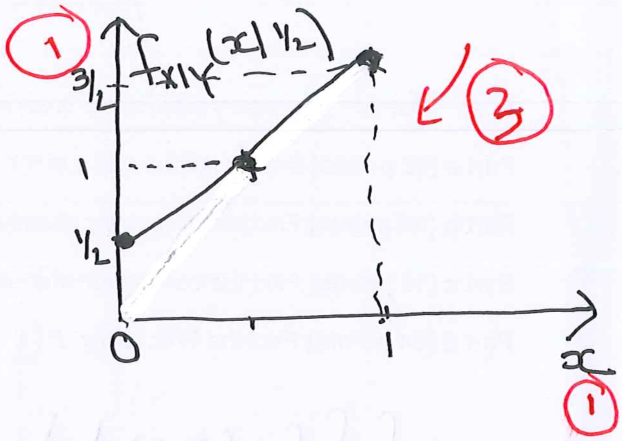
(c) $f_{X|Y}(x|y=1/2)$

2

$$= \frac{f_{X,Y}(x, y=1/2)}{f_Y(1/2)} = \frac{x+1/2}{1/2+1/2} = x+1/2$$

3

$f_{X|Y}(x|y=1/2) = x + 1/2, \ 0 \leq x \leq 1$



(d) $P(X \leq 1/2 \,|\, Y = 1/2)$

$$= \int_0^{1/2} f_{X|Y}(x|y=1/2)\,dx$$

1

$$= \int_0^{1/2} (x + 1/2)\,dx$$

$$\frac{x^2}{2} + \frac{1}{2}x \Big|_0^{1/2}$$

1

$$\left(\frac{1}{8} + \frac{1}{4}\right) - (0)$$

$$= 3/8$$

2